128 64 32 16 8 4 2 1

Given network: 192.168.52.0 /x

Network A / VLAN 9 | CS – 25 Hosts

2^n-2 = usable hosts, n = number of host bits

2^5-2 = 30 usable hosts

To find subnetmask, we subtract 32 to the number of host bits. 32 – 5 = 27

/27 = 255.255.255.224

Network Address: 192.168.52.0 255.255.255.224

First Usable Address: 192.168.52.1 255.255.255.224

Last Usable Address: 192.168.52.30 255.255.255.224

Broadcast Address: 192.168.52.31 255.255.255.224

Next Subnet: 192.168.52.32

Network B / VLAN 14 | IT – 10 Hosts

2^n-2 = usable hosts, n = number of host bits

2^4-2 = 14 usable hosts

To find subnetmask, we subtract 32 to the number of host bits. 32 – 4 = 28

/28 = 255.255.255.240

Network Address: 192.168.52.32 255.255.255.240

First Usable Address: 192.168.52.33 255.255.255.240

Last Usable Address: 192.168.52.46 255.255.255.240

Broadcast Address: 192.168.52.47 255.255.255.240

Next Subnet: 192.168.52.48 ← *ito na yung next available subnet kung may expansion na gagawin.*

Configuration

Hostname Configuration

SW100

Switch>en
Switch#conf t

Switch(config)#hostname SW100

SW200

Switch>en
Switch#conf t

Switch(config)#host SW200

R200

Router>en
Router#conf t
Router(config)#host R200

Access Ports Configuration on Switches

SW100

SW100>en
SW100#conf t
SW100(config)#int ran f0/1-3, f0/5
SW100(config-if-range)#switchport mode access

//Para maverify na naka access port na ang interface na cinonfigure, enter sh int f0/1 sw command sa privileged exec mode. I-check ang 'Administrative mode' kung naka 'static access' na.

SW100#show interface f0/1 switchport

SW200

SW200>en
SW200#conf t
SW200(config)#int ran f0/1-3
SW200(config-if-range)#switchport mode access

//Check ulit yung mga interfaces kung naapply ang configuration.

Creating VLANs 9 and 14 on Switches (May shortcut na rekta ang pag gawa ng vlans, pero ginawa ko ay nagcreate nalang ng mano mano)

SW100

```
SW100>en
SW100#conf t
SW100(config)#vlan 9
SW100(config-vlan)#name CS
SW100(config-vlan)#vlan 14
SW100(config-vlan)#name IT
```

SW200

```
SW200>en
SW200#conf t
SW200(config)#vlan 9
SW200(config-vlan)#name CS
SW200(config-vlan)#vlan 14
SW200(config-vlan)#name IT
```

Enrolling PCs to VLANs 9 and 14

SW100

```
SW100>en
SW100#conf t
SW100(config)#int ran f0/2-3, f0/5
SW100(config-if-range)#sw access vlan 14
SW100(config-if-range)#int f0/1
SW100(config-if)#sw acc vlan 9
```

SW200

```
SW200>en
SW200#conf t
SW200(config)#int ran f0/1-2
SW200(config-if-range)#sw acc vlan 9
SW200(config-if-range)#int f0/3
```

```
SW200(config-if)#switchport access vlan 14
```

Trunk Port Configuration on Switches SW100 and SW200

SW100

```
SW100>en
SW100#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW100(config)#int f0/4
SW100(config-if)#sw mode trunk
SW100(config-if)#sw trunk allowed vlan 9,14
```

//para maverify mo kung naging trunk nga, gamitin mo itong command na to ‘show interfaces trunk’

SW200

```
SW200>en
SW200#conf t
SW200(config)#int f0/4
SW200(config-if)#sw mode trunk
SW200(config-if)#sw trunk allowed vlan 9,14
```

Trunk Port Configuration on Switches SW200 to R200

SW200

```
SW200>en
SW200#conf t
SW200(config)#int g0/1
SW200(config-if)#sw mo tru
SW200(config-if)#sw tru allowed vlan 9,14
```

ROAS Configuration on R200

```
R200>en
R200#conf t
R200(config)#int g0/1
R200(config-if)#no shut
R200(config-if)#int g0/1.9
```

```
R200(config-subif)#encap dot1q 9
R200(config-subif)#ip address 192.168.52.30 255.255.255.224
R200(config-subif)#int g0/1.14
R200(config-subif)#encap dot1q 14
R200(config-subif)#ip add 192.168.52.46 255.255.255.240
```

//para makita mo ang configuration na nilagay sa subinterface, ‘show ip interface brief’